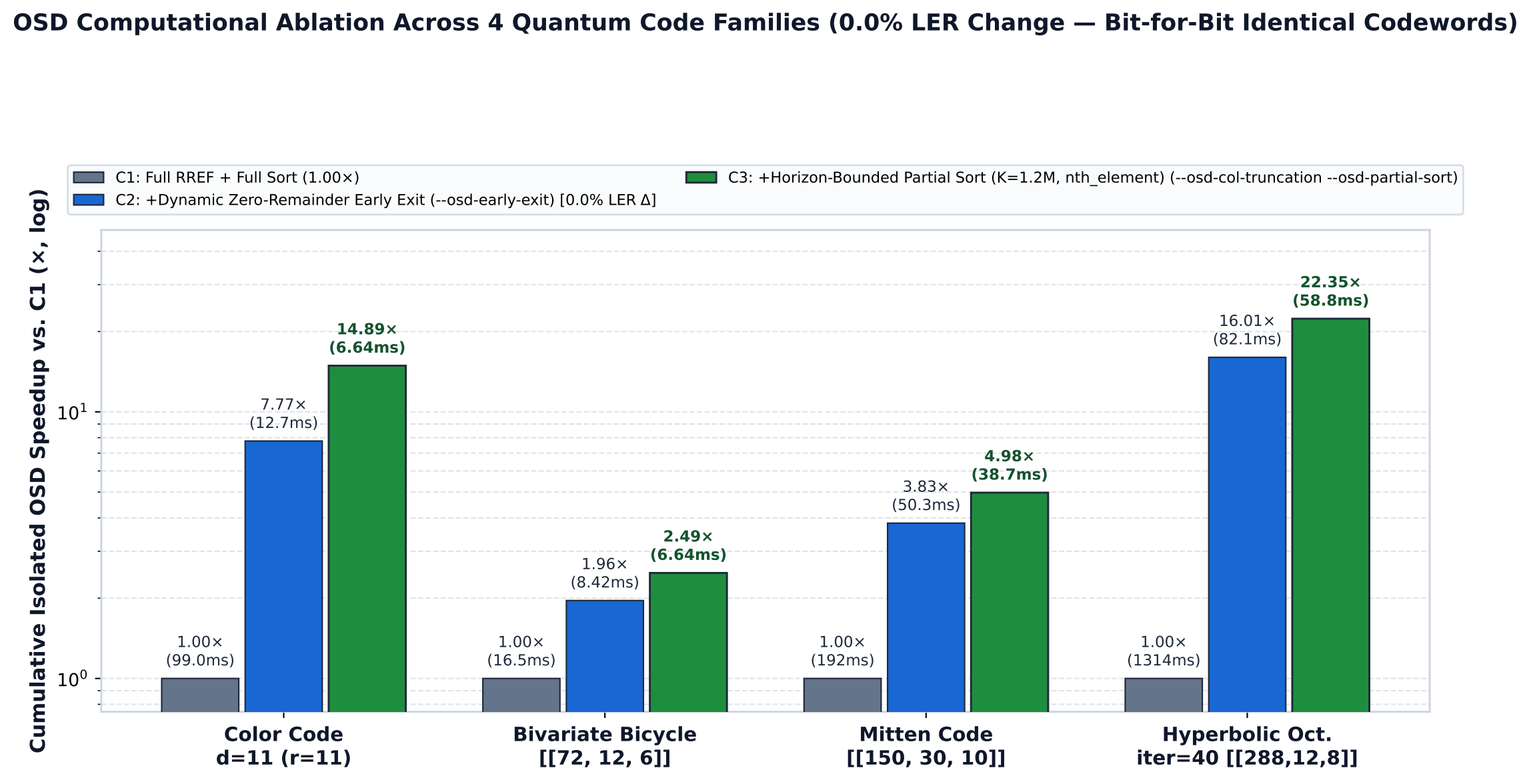
OSD Computational Ablation Across 4 Quantum Code Families (0.0% LER Change — Bit-for-Bit Identical Codewords)
C1: Full RREF + Full Sort (1.00×)
C2: +Dynamic Zero-Remainder Early Exit (--osd-early-exit) [0.0% LER Δ]
C3: +Horizon-Bounded Partial Sort (K=1.2M, nth_element) (--osd-col-truncation --osd-partial-sort)
Cumulative Isolated OSD Speedup vs. C1 (×, log)
$10^1$
$10^0$
1.00×
(99.0ms)
7.77×
(12.7ms)
14.89×
(6.64ms)
1.00×
(16.5ms)
1.96×
(8.42ms)
2.49×
(6.64ms)
1.00×
(192ms)
3.83×
(50.3ms)
4.98×
(38.7ms)
1.00×
(1314ms)
16.01×
(82.1ms)
22.35×
(58.8ms)
Color Code
d=11 (r=11)
Bivariate Bicycle
[[72, 12, 6]]
Mitten Code
[[150, 30, 10]]
Hyperbolic Oct.
iter=40 [[288,12,8]]